

■ Auto Garage Application — Vercel & Neon Deployment Guide

Prepared for: Team Lead & DevOps Administrator | Stack: Django 6.1, Vercel Serverless, Neon PostgreSQL, Cloudinary

1. Architectural Overview

- **Hosting & Compute:** Vercel (Serverless Python Runtime via `api/index.py` and `vercel.json`).
- **Production Database:** Neon PostgreSQL (Fully managed serverless Postgres with SSL connection support).
- **Static Assets:** Managed via whitenoise (Compressed Manifest Static Files Storage).
- **Media & Uploads:** Integrated with cloudinary & django-cloudinary-storage for invoice/vehicle images.

2. Prerequisites for Deployment

- **GitHub Access:** Push/Admin access to the repository.
- **Vercel Account:** Team or Individual Vercel Account linked to GitHub.
- **Neon Console:** Access to console.neon.tech to provision the Postgres DB.

3. Provisioning Neon PostgreSQL

1. Log into your Neon Console and click **Create Project** (e.g. `autogarage-prod-db`).
2. In the **Connection Details** dashboard, select **PostgreSQL** and copy your connection string:

```
postgres://<user>:<password>@<ep-hostname>.neon.tech/neondb?sslmode=require
```

4. Vercel Project Setup & Environment Variables

Import the GitHub repository into Vercel (Framework Preset: **Other**, Build Command: **bash build.sh**, Output Directory: **staticfiles**).

Add the following **Environment Variables** in Vercel Settings:

Environment Variable	Value / Description	Required
DATABASE_URL	postgres://<user>:<pass>@<ep-host>.neon.tech/neondb?sslmode=require	YES
SECRET_KEY	Secure random string (50+ characters)	YES
USE_LOCAL_DB	False	YES
CLOUDINARY_*	CLOUDINARY_CLOUD_NAME, API_KEY, API_SECRET	Optional

5. Database Migrations & Initial Owner Setup

Run migrations and create the initial Superuser account on Neon from your terminal:

```
# Windows PowerShell:
$env:DATABASE_URL="postgres://<user>:<pass>@<ep-host>.neon.tech/neondb?sslmode=require"
$env:USE_LOCAL_DB="False"
.\env\Scripts\python.exe manage.py migrate
.\env\Scripts\python.exe manage.py createsuperuser

# Linux / macOS:
export DATABASE_URL="postgres://<user>:<pass>@<ep-host>.neon.tech/neondb?sslmode=require"
export USE_LOCAL_DB="False"
python manage.py migrate
python manage.py createsuperuser
```

6. Post-Deployment Verification Checklist

- [✓] **Public Landing Page:** Visit <https://your-app.vercel.app/> — verify homepage loads cleanly.
- [✓] **Owner Portal Login:** Visit `/login/` and log in with your Superuser account.
- [✓] **Staff & Role Management:** Test staff registration (Service Advisor, Mechanic, Store Manager).
- [✓] **Job Cards & Invoices:** Verify creating a Job Card and generating an Invoice PDF.
- [✓] **Workshop Settings:** Upload workshop branding/logo in Garage Settings.